

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Collaborated with **FEOL/BEOL Integration** and **Yield Analysis** teams to diagnose root causes of yield deviations in semiconductor manufacturing, implementing corrective actions that improved process stability.
- Supported the import of new semiconductor technology into production fabs by working with **Technology Development** and local yield teams, ensuring smooth transition and adherence to targets.
- Managed product lifecycles in production environments, optimizing targets for **yield, device, and performance** to ensure successful product introduction and sustained operation.
- Analyzed **structured and unstructured data** from manufacturing processes to identify trends, deliver insights into process health, and inform data-driven decisions for improvement activities.
- Assessed quality and reliability risks associated with manufacturing materials and process changes, ensuring strict adherence to **Intel's standards** and regulatory compliance.
- Developed and refined **workflows and procedures** for continuous improvement programs, focusing on preventing yield excursions in wafer sort and line processes.

SKILLS

Semiconductor Manufacturing – FEOL/BEOL Integration, Yield Analysis, Process Optimization, Lithography, Advanced Patterning, Dry Etch, Wet Cleaning, Diffusion, Thin-Film Deposition, CMP, Technology Development, High Volume Manufacturing

Data Analysis & Insights – Data Mining, Statistical Analysis, Structured Data Analysis, Unstructured Data Analysis, Process Health Monitoring, Yield Excursion Prevention

Process Engineering & Quality – Root Cause Analysis, Corrective Action Implementation, Product Maintenance, Quality Risk Assessment, Reliability Risk Assessment, Continuous Improvement

Collaboration & Communication – Cross-functional Team Collaboration, Stakeholder Management, Technical Presentations, Problem-Solving, Decision-Making

Core Technical Skills – Java, Python, SQL, Spring Boot, REST APIs, AWS, Docker, Linux, Git, Jenkins, Jira, Power BI, Tableau, R, scikit-learn

PROJECTS

Semiconductor Yield Optimization Initiative (HCL Technologies)

- Analyzed historical manufacturing data using **Python** and **SQL** to identify key parameters affecting yield in a critical production line, contributing to a focused improvement effort.
- Developed automated data validation scripts in **Python** for incoming process logs, reducing manual data cleaning time by approximately **40%** and improving data integrity for analysis.
- Collaborated with operations teams to interpret data insights and recommend process adjustments, supporting the implementation of changes that stabilized yield by approximately **15%** over a quarter.

Production Data Analysis Dashboard (HCL Technologies)

- Designed and built an interactive dashboard using **Power BI** to visualize key production metrics and trends for a financial services operations team, enabling faster identification of performance bottlenecks.
- Implemented complex **DAX** measures to calculate critical KPIs, such as transaction processing rates and error frequencies, providing actionable insights to business analysts.
- Integrated data from multiple sources using **Power Query**, transforming and modelling it to support comprehensive reporting and ad-hoc analysis for stakeholders.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)